Ardan Labs

Ardan Labs specializes in high-performance software engineering solutions using Go, Rust, Docker, and Kubernetes, with deep expertise in APIs, AI implementations, distributed systems and blockchain.

- Official website: https://www.ardanlabs.com/
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Who we are

Ardan Labs is a software engineering firm founded around the vision of **Bill Kennedy** (Managing Partner, co-author of *Go in Action* and founder of the GoBridge Foundation), a worldwide reference in **software architecture, Go, and Rust**. The firm specializes in high-performance software engineering solutions using **Go, Rust, Docker, and Kubernetes**, with deep expertise in APIs, platforms, AI, and databases (SQL & NoSQL).

Delivery spans **AWS, GCP, and Azure**, leveraging advanced DevOps, CI/CD automation, and distributed systems design. Ardan Labs takes full ownership of complex projects — debugging legacy code, refactoring, modernizing monolithic applications, and building platforms to unlock data for AI — speeding up time-to-market without sacrificing performance.

Core specialties

- Software Development
- Staff Augmentation
- Consulting
- Training

Skills & Technologies

Expert Technologies

- *Programming:* Go & Rust
- *DevOps:* Docker, Kubernetes, and cloud-native (cloudless) technologies
- *Databases:* SQL & NoSQL

Complimentary Technologies

- *Web & Frontend:* HTML/CSS and React
- *Web Servers & Reverse Proxies:* Nginx and Caddy
- *CI/CD & Infrastructure as Code:* Terraform, GitHub Actions, Vault, and Bazel
- *APIs & Authentication:* REST, gRPC, and JWT
- *AI/ML Infrastructure:* AI servers (TGI, TEI, vLLM, llama.cpp, etc.)
- *Observability & Monitoring:* Grafana, Prometheus, Datadog, and OTEL
- *Access Control & Policy Enforcement:* Cerbo (OpenTelemetry)

AI Solutions

- **Reducing Cloud Costs** — architect local inference on your own hardware to flip from variable OpEx (per-token costs) to stable CapEx.
- **Data Sovereignty** — air-gapped AI solutions for Finance and Healthcare, where proprietary data never leaves your network.
- **Python-to-Production Gap** — bridge research-in-Python to production-at-scale in Go or C++ with high-concurrency systems that don't sacrifice performance.
- **Latency-Critical UX** — hardware-accelerated inference (Metal, CUDA, Vulkan) for sub-second response times on edge and desktop apps.
- **Solid RAG Pipelines** — rigorous Retrieval-Augmented Generation systems that steer models to speak only from verified corporate data.
- **Model Fine Tuning** — tighter interactions with LLMs when context grows past ~50% and accuracy or speed starts to degrade.

Blockchain Solutions

- **Governance Platforms** — token-weighted voting, delegation, timelocked execution, and treasury management fully on-chain.
- **Permissioned Trading Environments** — token transfer hooks and KYC implementation to bring real-world assets and regulated products on-chain.
- **Smart Contract Development, Auditing & Protocol Design** — Solana and Ethereum, including trading, governance, infrastructure primitives, and security reviews.
- **Indexing and Trading Infrastructure** — production-grade ingestion of program logs, account state changes, and transaction metadata on Solana.

Problems we solve

1. **Improving developer productivity with smarter tooling** — optimized CI/CD pipelines, automated repetitive tasks, and powerful dev tools that move teams faster without compromising quality.
2. **Eliminating technical debt** — refactoring legacy codebases with best practices and clean architecture so systems stay scalable and easy to extend.
3. **Eliminating deployment and infrastructure bottlenecks** — seamless cloud deployment (AWS/GCP/Azure) with DevOps, CI/CD automation, and containerization for scale and reliability.
4. **Scaling APIs and platforms without downtime** — event-driven architecture, distributed systems, and efficient SQL/NoSQL data management for massive traffic with low latency.
5. **Bringing structure to disorganized DevOps & CI/CD** — pipelines, automated testing, and infrastructure as code that make deployments smooth and predictable.

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*Thanks to Ardan Labs for sponsoring **The 500MB Club** — their expertise in Go, Rust, and high-performance architecture resonates directly with the spirit of this challenge.*